

```
'''Assignmnet No: 9 Data Visualization II '''
```

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

```
df = pd.read_csv('titanic.csv')
df
```

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	
..	...	...	...	
886	887	0	2	
887	888	1	1	
888	889	0	3	
889	890	1	1	
890	891	0	3	

SibSp	\	Name	Sex	Age
0		Braund, Mr. Owen Harris	male	22.0
1				
1		Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0
1				
2		Heikkinen, Miss. Laina	female	26.0
0				
3		Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0
1				
4		Allen, Mr. William Henry	male	35.0
0				
..		...	...	...
...				
886		Montvila, Rev. Juozas	male	27.0
0				
887		Graham, Miss. Margaret Edith	female	19.0
0				
888		Johnston, Miss. Catherine Helen "Carrie"	female	NaN
1				
889		Behr, Mr. Karl Howell	male	26.0
0				
890		Dooley, Mr. Patrick	male	32.0
0				

Parch	Ticket	Fare	Cabin	Embarked
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0	0	A/5	21171	7.2500	NaN	S
1	0	PC	17599	71.2833	C85	C
2	0	STON/O2.	3101282	7.9250	NaN	S
3	0		113803	53.1000	C123	S
4	0		373450	8.0500	NaN	S
...	...		...	...	...	...
886	0		211536	13.0000	NaN	S
887	0		112053	30.0000	B42	S
888	2	W./C.	6607	23.4500	NaN	S
889	0		111369	30.0000	C148	C
890	0		370376	7.7500	NaN	Q

[891 rows x 12 columns]

```
cols = df.columns
cols
```

```
Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age',
      'SibSp',
      'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'],
      dtype='object')
```

```
df.info()
```

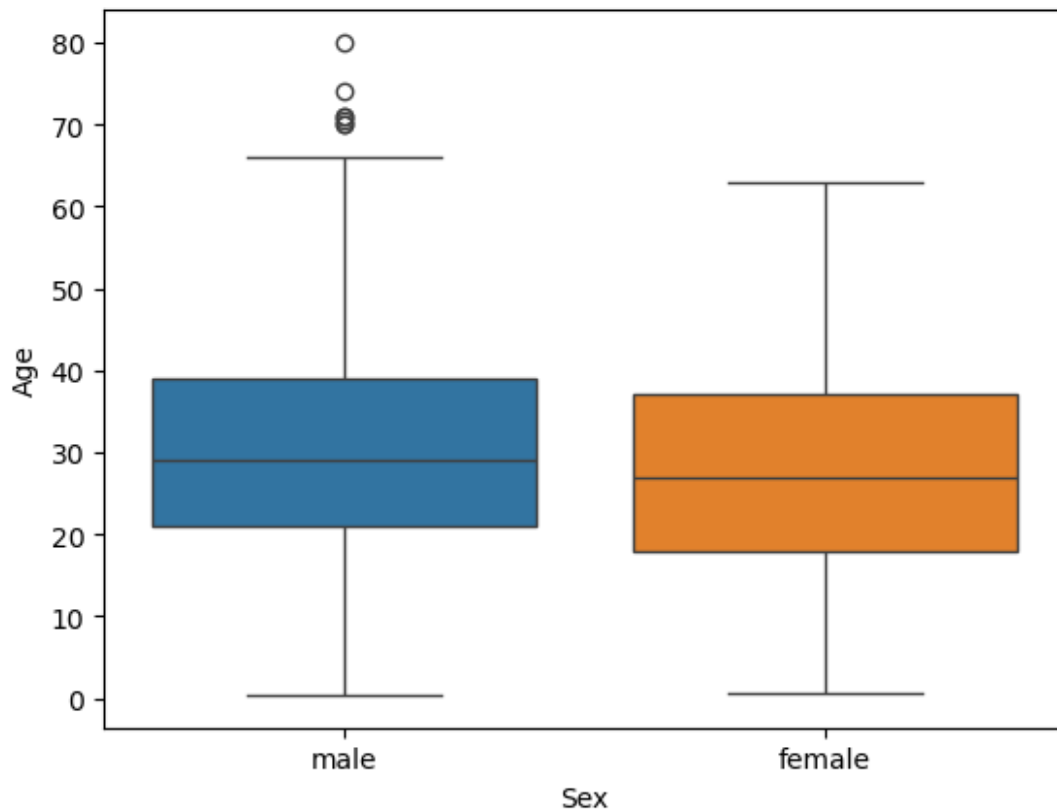
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
5   Age          714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    object
9   Fare         891 non-null    float64
10  Cabin        204 non-null    object
11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
df.isnull().sum()
```

PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0

```
Age          177
SibSp         0
Parch         0
Ticket        0
Fare          0
Cabin        687
Embarked      2
dtype: int64
```

```
sns.boxplot(x='Sex', y='Age', hue='Sex', data=df)
plt.show()
```



```
sns.boxplot(x='Sex', y='Age', hue='Survived', data=df)
plt.show()
```

